We will denote the average with respect to the canonical probability density by Av_β. Using (6.4), we can compute the average of the function p^2 as

$$\begin{aligned}\mathrm{Av}_\beta(p^2) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p^2 \rho_\beta \mathrm{d}q\mathrm{d}p \\ &= \frac{\int_{-\infty}^{\infty} p^2 \exp(-\beta m^{-1}p^2/2)\mathrm{d}p}{\int_{-\infty}^{\infty} \exp(-\beta m^{-1}p^2/2)\mathrm{d}p},\end{aligned}$$

where we have used the fact that the exponential of H separates into a product of p and q dependent factors. We then integrate by parts to obtain

$$\mathrm{Av}_\beta(p^2) = m\beta^{-1},$$

which means that

$$\mathrm{T} = \mathrm{Av}_\beta\left(\frac{p^2}{m\mathrm{k_B}}\right).$$

Thus the temperature can be related to the canonical average of the kinetic energy.

If we consider a $\mathrm{N_d} = \mathrm{N_c}$ degree of freedom system with kinetic energy $K = \sum_{i=1}^{\mathrm{N_c}} p_i^2/(2m_i)$ and potential energy $U(q_1, q_2, \ldots, q_{\mathrm{N_c}})$, then we may decompose the unnormalized canonical distribution as

$$\begin{aligned}\exp\left(-\beta[K(\boldsymbol{p}) + U(\boldsymbol{q})]\right) &= \exp\left(-\beta K(\boldsymbol{p})\right)\exp\left(-\beta U(\boldsymbol{q})\right) \\ &= \exp\left(-\beta U(\boldsymbol{q})\right)\prod_{i=1}^{\mathrm{N_c}}\exp\left(-\beta\frac{p_i^2}{2m_i}\right).\end{aligned}$$

Then we again find, by the integration argument given above, that

$$\mathrm{Av}_\beta(m_i^{-1}\mathrm{k_B}^{-1}p_i^2) = \mathrm{T}.$$

Thus the average kinetic energy of any individual particle is directly related to the temperature, and thus also

$$\mathrm{Av}_\beta(2K) = \mathrm{N_c}\mathrm{k_B}\mathrm{T}.$$

Very often, this is taken to be the definition of temperature in practical settings. However, since the temperature relation is derived by integration with respect to